VIDYAPEETH

BATCH CODE: 12-AN181MA 2026

SUBJECT NAME: PHYSICS

CHAPTER NAME:
UNITS AND MEASURMENT

Lecture No. 10

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Today's Goal

Subtopic

SIGNIFICANT FIGURES.

Subtopic

Subtopic

ACCURACY & PRECISION

a) 1.45
b) 1.499
c) 1.4217
d) 5.79815

a) 1.439 cm
b) 1.48 cm
c) 1.4399 cm
d) 1.42 cm

$l = 1.44$ cm

Least Count
1mm
0.1 mm
0.01 mm

Accuracy → closest value

Precision ⟹ Most Detailed Value
(Deci. के बाद सबसे ज्यादा नं०)

Significant Figures :- RULES :-

p. 000572

Rule ① All Non-Zero digits are SF

Eg:- 1357
SF = 4

Rule ② Zeroes appearing b/w two non-zero digits are significant, no matter where the decimal is.

Eg:- 13057 → SF = 5
13.057 → SF = 5
2005 → SF = 4
2.005
20.05
200.5
SF = 4

R③ In Nos. less than 1, zeroes right to the decimal but left to the first non-zero digit are not significant

Eg:- 0.0057 SF = 2
0.00577 SF = 3

0.00507 SF = 3
Not SF
SF

Rule ④ Trailing zeroes in a no. "without Decimal" are Not significant

Eg: 420 SF = 2

4200 SF = 2

4.200000 = SF 2

Rule ⑤ In a no, zeroes right to the last non-zero digit are SIGNIFICANT.

Eg:- 7.3000

SF = 5

7.302000

SF = 7.

Rule 6

★ **Point of Confusion:-**

2.30 m SF = 3

230 cm SF = 2

2300 mm SF = 2

confusion

NOTE:- i) Count of SF doesnot changes on change in unit.

ii) Always write in Scientific Notation.

Solution

$a \times 10^b$

$a < 10$

$10^{\cdots}$

$2.30\ m = 2.30 \times 10^0\ m$

$230\ cm = 2.30 \times 10^2\ cm$

$2300\ mm = 2.30 \times 10^3\ mm$

SF = 3

$= 2.30 \times 10^{-3}\ km$

SF = 3

Rule 7

Any exact no. appearing in a Mathematical formula has "INFINTE SFs".

Eg. i) Perimeter of square = $4 \times a$

$4 \rightarrow 4.000000000\ldots$

$4 \rightarrow SF = \infty$

ii) $\pi \rightarrow SF = \infty$

iii) $C = 2\pi r$

$SF = \infty$

Read NCERT also

$2.30\,m \rightarrow$

$\rightarrow \underbrace{2.30}\times 10^{3}\ mm$

$\rightarrow \underbrace{2.30}\times 10^{2}\ cm$

$\rightarrow \underbrace{2.30}\times 10^{-3}\ km$ $\rightarrow$ X

SF = 3

7.00

7.01

Rule 6

★★★★★

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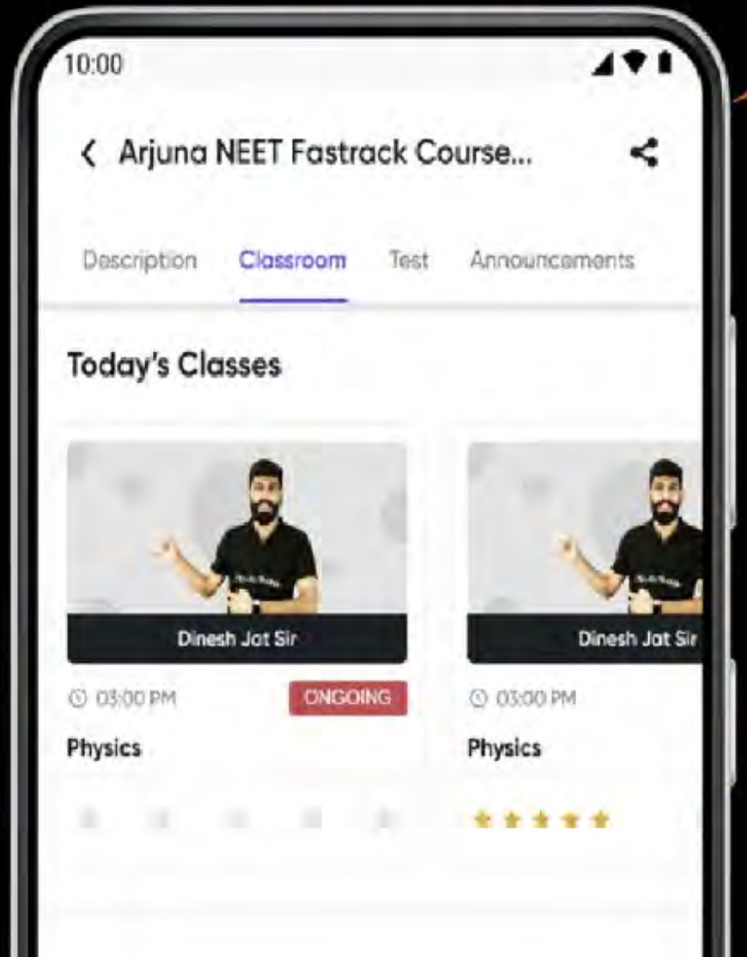
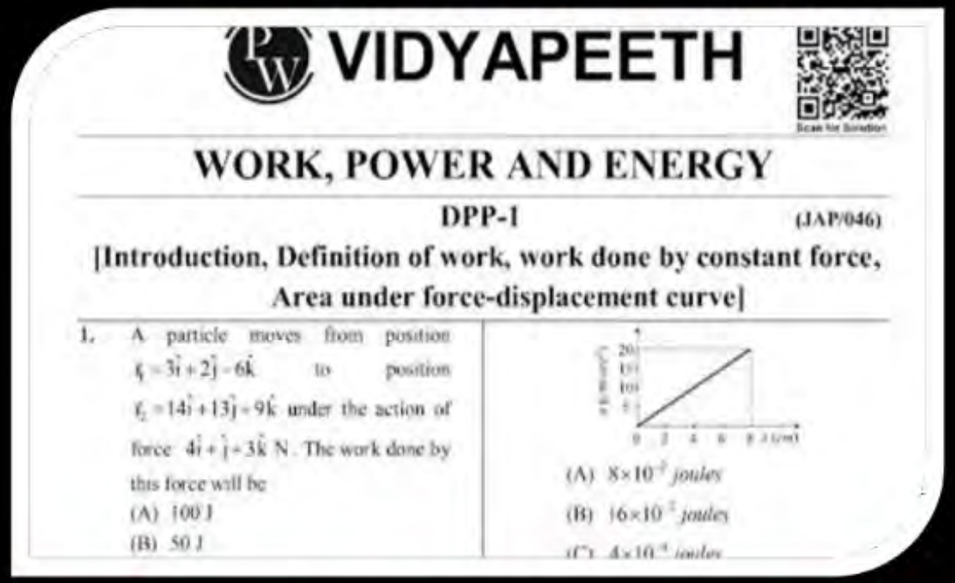


Solve the DPP and check Solution



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